

Cisco Packet Tracer Project #03

Inter-VLAN Routing — Router-on-a-Stick

1. Project Overview

This project demonstrates Inter-VLAN Routing using Router-on-a-Stick. VLAN 10 (IT) and VLAN 20 (Finance) use separate IP networks and communicate through router subinterfaces over an 802.1Q trunk.

2. Objectives

- Configure VLAN 10 and VLAN 20.
- Configure access ports and an 802.1Q trunk.
- Create router subinterfaces for each VLAN.
- Configure default gateways.
- Verify and troubleshoot inter-VLAN connectivity.

3. Devices and Addressing

Device	Interface	VLAN	IP Address	Gateway
R1	G0/0.10	10	192.168.10.1	N/A
R1	G0/0.20	20	192.168.20.1	N/A
PC-01	NIC	10	192.168.10.10	192.168.10.1
PC-02	NIC	10	192.168.10.11	192.168.10.1
PC-03	NIC	10	192.168.10.12	192.168.10.1
PC-04	NIC	20	192.168.20.10	192.168.20.1
PC-05	NIC	20	192.168.20.11	192.168.20.1
PC-06	NIC	20	192.168.20.12	192.168.20.1

4. Router-on-a-Stick Configuration

R1 G0/0.10 uses 802.1Q VLAN 10 with gateway 192.168.10.1/24. R1 G0/0.20 uses 802.1Q VLAN 20 with gateway 192.168.20.1/24. The physical G0/0 interface was enabled with no shutdown.

5. Verification and Testing

SW1 VLAN verification confirmed Fa0/1–Fa0/3 in VLAN 10 and Fa0/4–Fa0/6 in VLAN 20. The trunk on Fa0/24 became operational after R1 G0/0 was enabled. Inter-VLAN ping testing between PC-01 (192.168.10.10) and PC-04 (192.168.20.10) completed with 0% packet loss.

6. Troubleshooting

Fa0/24 initially showed notconnect and Operational Mode: down even though its administrative mode was trunk. The router-side G0/0 interface was not active. Applying no shutdown to R1 G0/0 restored the physical link and allowed the trunk to operate.

7. Key Concepts Learned

VLAN segmentation, access ports, 802.1Q trunking, Router-on-a-Stick, subinterfaces, default gateways, IPv4 addressing, Layer 2 and Layer 3 routing, ICMP testing, and structured troubleshooting.

8. Conclusion

The completed topology successfully enabled communication between separate VLANs through R1 using Router-on-a-Stick. This project extends the Layer 2 VLAN concepts from Project #02 into Layer 3 Inter-VLAN Routing.